

WATERTON PARK GATE, AB, Canada

WMO: 711540

Lat: 49.1311N

Lon: 113.8086W

Elev: 1289

StdP: 86.77

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.9	-24.7	-32.4	0.2	-28.7	-27.9	0.3	-24.4	20.0	4.6	18.7	2.0	1.4	40	0.767

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	29.4	15.0	27.6	14.6	25.8	14.1	17.0	25.3	16.0	24.9	15.0	23.8	4.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.0	11.7	20.8	12.6	10.7	19.1	11.4	9.8	17.8	53.0	25.2	49.5	25.0	46.4	23.9	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	14.3	12.7	-35.3	32.3	5.2	2.0	-39.1	33.7	-42.1	34.8	-45.1	35.9	-48.9	37.4		
			-35.4	19.2	5.2	1.0	-39.2	20.0	-42.2	20.6	-45.1	21.1	-48.9	21.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-4.2	-4.1	-1.1	4.0	9.0	12.7	16.6	15.8	11.7	5.8	0.0	-4.0
	DBStd	9.82	8.83	8.30	7.94	5.35	4.64	3.55	3.09	3.40	4.78	5.81	7.71	8.06
	HDD10.0	2384	441	396	344	188	75	13	1	3	36	150	303	435
	HDD18.3	4818	698	629	602	430	291	172	71	91	203	388	550	693
	CDD10.0	643	1	0	1	8	43	94	205	182	86	21	3	0
	CDD18.3	36	0	0	0	0	1	3	17	13	3	0	0	0
	CDH23.3	1014	0	0	0	1	19	64	432	388	107	3	0	0
	CDH26.7	254	0	0	0	0	1	13	123	100	17	1	0	0
Wind	WSAvg	5.5	7.5	6.4	5.7	5.1	4.6	4.5	3.7	3.5	4.3	6.0	7.0	7.5
Precipitation	PrecAvg	849	69	56	65	80	96	126	57	58	66	71	75	60
	PrecMax	1177	175	103	97	143	230	291	190	148	148	169	136	104
	PrecMin	647	30	18	22	38	21	26	0	15	7	29	39	38
	PrecStd	131	30	18	19	22	47	72	41	35	33	30	25	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.2	11.9	14.9	20.7	25.7	28.9	32.1	31.6	28.8	22.4	16.2	10.9
		MCWB	4.9	4.1	6.1	9.3	12.3	15.9	15.9	14.8	13.8	10.4	6.9	4.8
	2%	DB	8.9	8.9	12.0	17.7	22.9	24.9	29.7	29.2	26.3	18.8	13.1	8.2
		MCWB	3.0	2.9	5.1	7.7	11.3	14.1	15.5	14.5	13.1	9.5	5.8	2.8
	5%	DB	6.7	6.9	9.9	14.8	20.2	22.8	27.8	27.4	24.1	16.1	10.7	6.2
		MCWB	2.2	1.8	3.9	6.4	10.2	13.2	15.1	14.2	12.2	8.0	4.9	1.9
	10%	DB	5.3	5.2	8.0	12.0	17.7	20.6	25.7	25.5	21.2	13.8	8.7	4.6
		MCWB	1.4	1.0	2.9	5.1	9.0	11.9	14.9	14.0	11.2	6.9	3.7	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.7	4.8	7.0	9.9	13.4	17.5	18.9	17.6	15.7	11.8	8.5	5.4
		MCDB	10.7	9.9	13.3	19.6	23.4	25.1	26.6	25.9	24.8	19.9	14.1	9.8
	2%	WB	3.8	3.6	5.6	8.1	12.0	15.3	17.5	16.4	14.1	9.9	6.8	3.7
		MCDB	7.5	7.8	10.7	16.5	21.0	23.4	25.6	25.3	23.7	17.9	12.4	6.9
	5%	WB	2.6	2.3	4.4	6.7	10.8	13.9	16.5	15.4	12.9	8.6	5.2	2.4
		MCDB	6.0	6.5	9.3	13.8	19.2	21.4	25.1	24.6	22.3	15.3	10.2	5.5
	10%	WB	1.6	1.1	3.2	5.5	9.6	12.6	15.5	14.5	11.6	7.3	3.9	1.1
		MCDB	5.1	5.0	7.6	11.6	16.8	19.0	24.2	23.6	20.2	13.2	8.0	4.4
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.4	9.9	10.3	11.2	11.6	15.2	15.8	13.2	9.8	8.1	7.8
		MCDBR	8.3	8.9	10.7	14.7	15.8	16.4	19.9	20.8	19.2	13.0	9.1	7.4
		MCWBR	5.4	5.6	6.1	7.6	7.5	8.2	8.7	8.9	8.8	6.8	5.3	5.0
	5% WB	MCDBR	7.8	8.9	10.4	13.7	14.3	15.4	17.5	18.1	18.0	13.1	8.8	7.7
		MCWBR	5.5	6.1	6.5	7.4	7.1	8.3	9.0	8.8	8.7	7.2	5.6	5.6
Clear-Sky Solar Irradiance	taub	0.238	0.250	0.276	0.337	0.342	0.334	0.331	0.346	0.315	0.281	0.255	0.229	
	taud	2.384	2.367	2.365	2.252	2.374	2.449	2.470	2.420	2.486	2.561	2.498	2.401	
	Ebn at Noon	851	920	946	909	917	925	922	891	891	876	827	816	
	Edh at Noon	63	82	99	124	115	108	104	105	88	68	57	54	
All-Sky Solar Radiation	RadAvg	1.19	2.17	3.34	4.67	5.59	6.18	7.03	5.68	4.01	2.44	1.31	0.94	
	RadStd	0.10	0.17	0.31	0.34	0.48	0.50	0.45	0.38	0.36	0.25	0.11	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page